Figure 12b · One epoch, full data

RTX 6000 Ada · 32 test cases per family · Weight EMA

Fast paths and both decoder paths: measured timings. Other native screens: shared timing proxy.

Stokes
T = 20
Tanaka
T = 200
Benjamin-Feir
T = 200
JONSWAP / TMA
T = 20
Median final surface error (%)
Single-rollout runtime (seconds)
Learning rate 1e−5
0.03444%
Control
0.02823%
Reflection augmentation
0.02817%
Multiplier skip
0.02762%
Momentum × ¼
0.02739%
Momentum × ½
0.02728%
Mode balancing × ⅓
0.02682%
Longer LR plateau
Mode balancing × ⅓
1.182%
Learning rate 1e−5
0.953%
Reflection augmentation
0.935%
Control
0.7349%
Multiplier skip
0.723%
Momentum × ¼
0.7196%
Learning rate 4e−5
0.6694%
Momentum × ½
Nested · FP32 training gate · fast
0.00225%
Nested · zero inner · fast
0.002172%
Batch 32 · HD × 4 · modes ≤128 · fast
0.001864%
Nested · FP32 · B16 · fast
0.001751%
Nested · FP32 · B16 · FFT fusion trial
0.00175%
Nested · FP32 · B16 · batched FFTs
0.00175%
Batch 32 · HD × 4 · xi norm · grid512
0.001737%
Batch 32 · HD × 4 · xi norm · fast
0.001736%
Nested · FP32 · B16 · seed 1 · fast
0.001735%
Nested · FP32 · B32 · seed 1 · fast
0.001542%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.001525%
Nested · seed 1 · fast
0.001498%
Nested · surface modulation · fast
0.001422%
Band ≤128 · 4 nested branches · fast
0.001262%
Nested · FFT fusion trial
0.001258%
CS-field decoder · optimized
3.519e-05%
CS-field decoder · native
3.519e-05%
M4
M3 decoder: 4.27 → 2.57 s (-39.8%)
10⁻⁶
0.0
2.5
5.0
7.5
0.04868%
Band ≤128 · 4 nested branches · fast
0.04712%
Nested · FFT fusion trial
0.04712%
Nested · surface modulation · fast
0.0427%
Nested · FP32 · B16 · FFT fusion trial
0.03888%
Nested · FP32 · B16 · batched FFTs
0.03888%
Nested · FP32 · B16 · fast
0.03887%
Batch 32 · HD × 4 · xi norm · fast
0.03882%
Batch 32 · HD × 4 · xi norm · grid512
0.03881%
Nested · FP32 · B32 · seed 1 · fast
0.03605%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.03578%
Nested · FP32 · B16 · seed 1 · fast
0.03183%
Batch 32 · HD × 4 · modes ≤128 · fast
0.02655%
Nested · seed 1 · fast
0.02655%
Nested · FP32 training gate · fast
0.02544%
CS-field decoder · native
0.002576%
CS-field decoder · optimized
0.002576%
M4
M3 decoder: 41.86 → 25.27 s (-39.6%)
10⁻³
0
25
50
75
0.2605%
Batch 128 · seed 1 · exact G₀
0.2178%
Batch 128 · seed 1 · fast
0.2177%
Batch 64 · Hadamard flow probes · fast
0.2129%
Batch 32 · HD × 4 · xi norm · fast
0.1875%
Batch 32 · HD × 4 · xi norm · grid512
0.1875%
LR 8e−5 + batch 128 · fast
0.1673%
Batch 64 · softened curvature · fast
0.1598%
Batch 64 · Hadamard × 16 · fast
0.1595%
Batch 64 · Hadamard × 4 · fast
0.1593%
Batch 64 · seed 1 · fast
0.1592%
LR 8e−5 + batch 64 · fast
0.1555%
Batch 32 · HD × 4 · xi norm · seed 1 · fast
0.1542%
Batch 32 · HD × 4 · modes ≤128 · fast
0.154%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.1467%
Batch 64 · Hadamard × 4 · seed 1 · fast
0.1421%
Batch 32 · Hadamard × 4 · fast
0.1335%
Batch 32 · Hadamard × 4 · seed 1 · fast
0.1223%
Nested · FP32 · B32 · seed 1 · fast
0.09684%
Nested · seed 1 · fast
0.08975%
Nested · zero inner · fast
0.08908%
Nested · FP32 · B16 · seed 1 · fast
0.08197%
Nested · surface modulation · fast
0.07705%
Nested · FP32 training gate · fast
0.07353%
Nested · FFT fusion trial
0.06596%
Band ≤128 · 4 nested branches · fast
0.06595%
Nested · FP32 · B16 · FFT fusion trial
0.06516%
Nested · FP32 · B16 · fast
0.06516%
CS-field decoder · optimized
0.005384%
CS-field decoder · native
0.005384%
M4
M3 decoder: 41.85 → 25.29 s (-39.6%)
10⁻²
0
25
50
75
0.1532%
LR 8e−5 + batch 128 · fast
0.1434%
Batch 64 · Hadamard flow probes · fast
0.1434%
Batch 64 · Hadamard × 4 · seed 1 · fast
0.1254%
Batch 32 · Hadamard × 4 · seed 1 · fast
0.1239%
Batch 64 · seed 1 · fast
0.1192%
Batch 64 · Hadamard × 4 · fast
0.1189%
Batch 128 · seed 1 · exact G₀
0.1178%
Batch 128 · seed 1 · fast
0.1178%
LR 8e−5 + batch 64 · fast
0.1173%
Batch 64 · softened curvature · fast
0.1156%
Batch 64 · Hadamard × 16 · fast
0.1035%
Batch 32 · HD × 4 · xi norm · seed 1 · fast
0.1014%
Batch 32 · HD × 4 · xi norm · grid512
0.09909%
Batch 32 · HD × 4 · xi norm · fast
0.09909%
Batch 32 · Hadamard × 4 · fast
0.08944%
Batch 32 · HD × 4 · ≤128 · seed 1 · fast
0.06454%
Batch 32 · HD × 4 · modes ≤128 · fast
0.06239%
Nested · zero inner · fast
0.04708%
Nested · FP32 · B32 · seed 1 · fast
0.04691%
Nested · seed 1 · fast
0.04577%
Nested · FFT fusion trial
0.04324%
Band ≤128 · 4 nested branches · fast
0.04324%
Nested · FP32 training gate · fast
0.04222%
Nested · surface modulation · fast
0.03982%
Nested · FP32 · B16 · seed 1 · fast
0.03495%
Nested · FP32 · B16 · fast
0.02971%
Nested · FP32 · B16 · FFT fusion trial
0.02971%
CS-field decoder · optimized
0.0008379%
CS-field decoder · native
0.0008379%
M4
M3 decoder: 4.27 → 2.57 s (-39.8%)
10⁻³
10⁻⁴
0.0
2.5
5.0
7.5